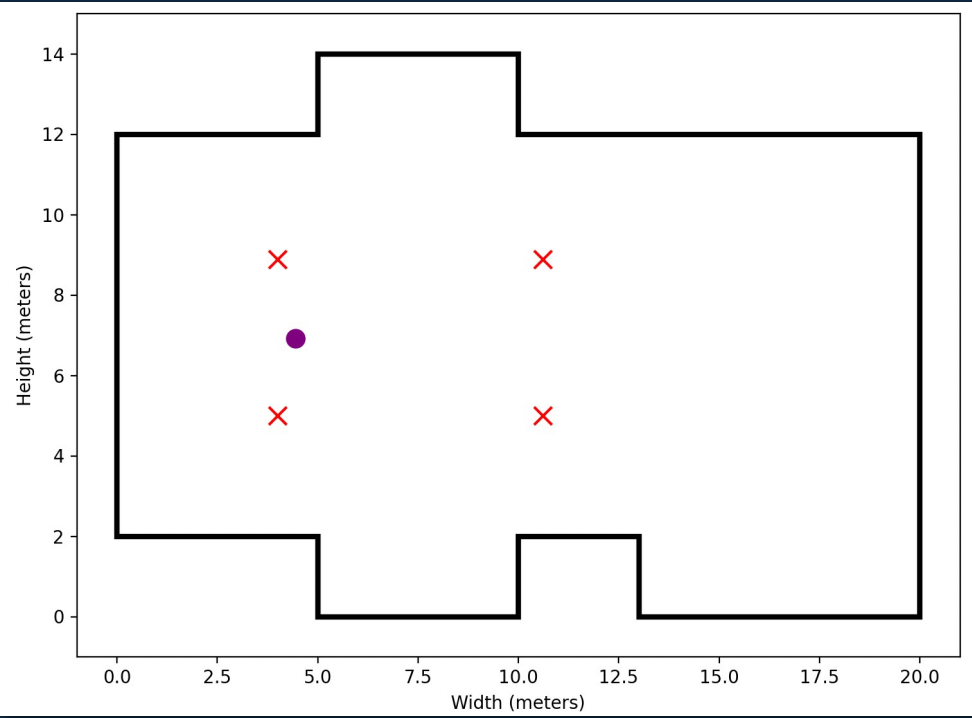


Algorithmic Framework for Sound Localization in Polyphonic Noisy Environments

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A **python library** is introduced for **localization of sound sources** in noisy, polyphonic environments



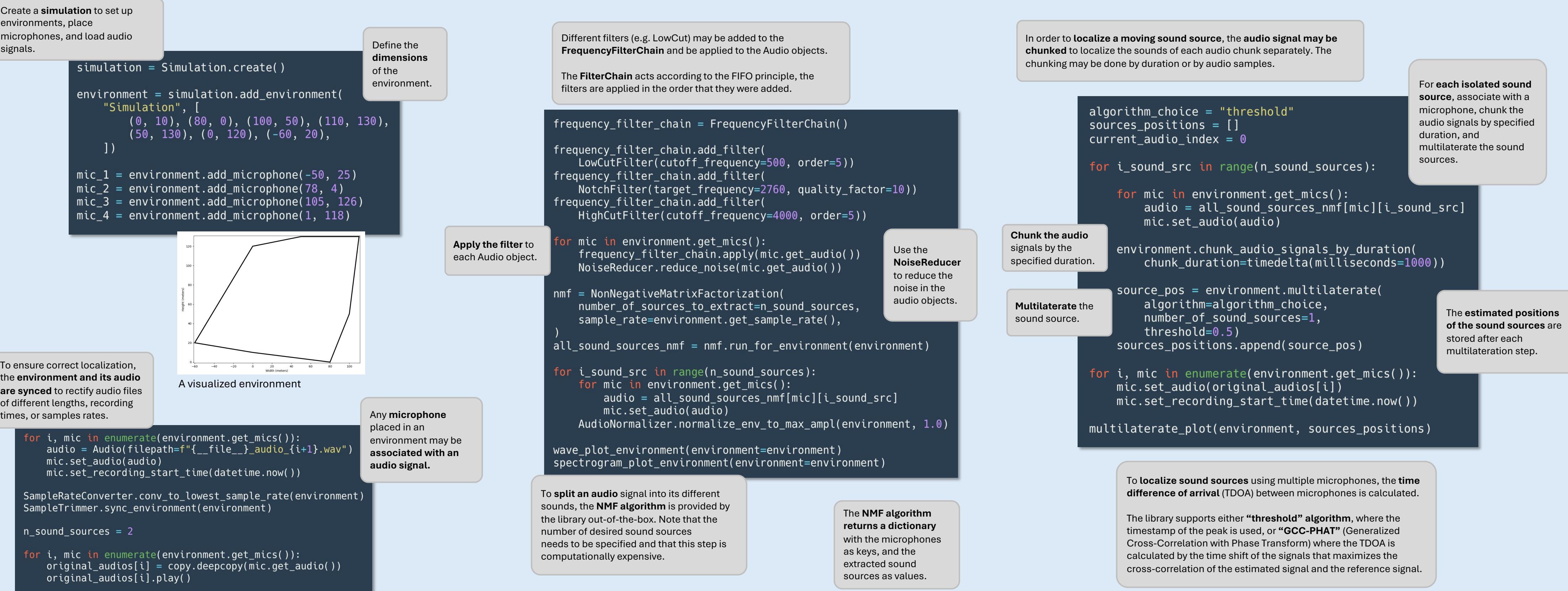
A Simulation
Containing a 2D environment, four distinct microphones, and a localized sound source.

Pipeline

Environment 01

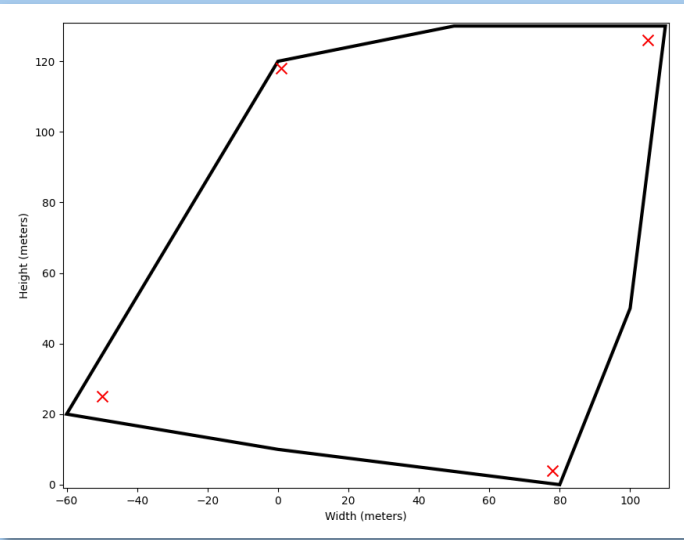
Pre-processing 02

Localization 03

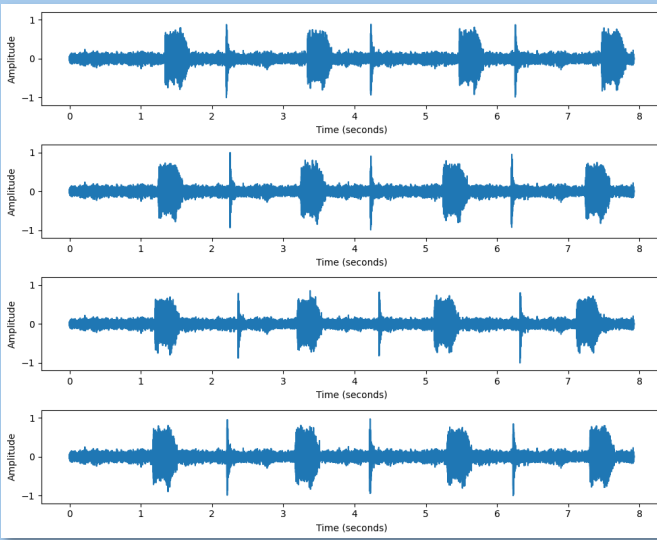


Results

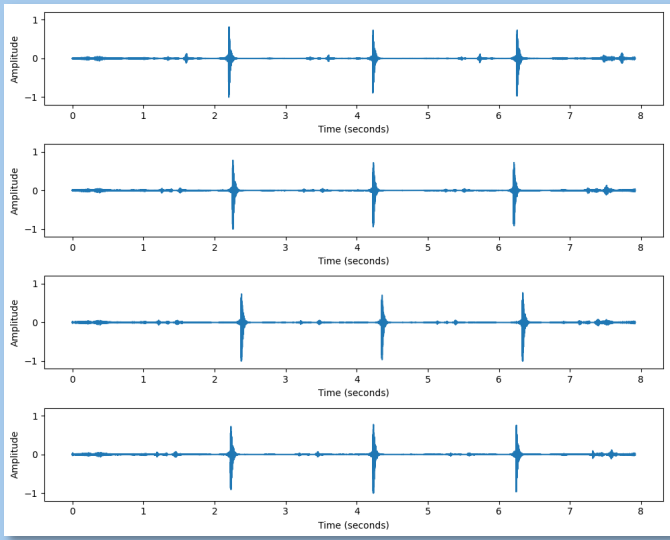
Environment with 4 Microphones



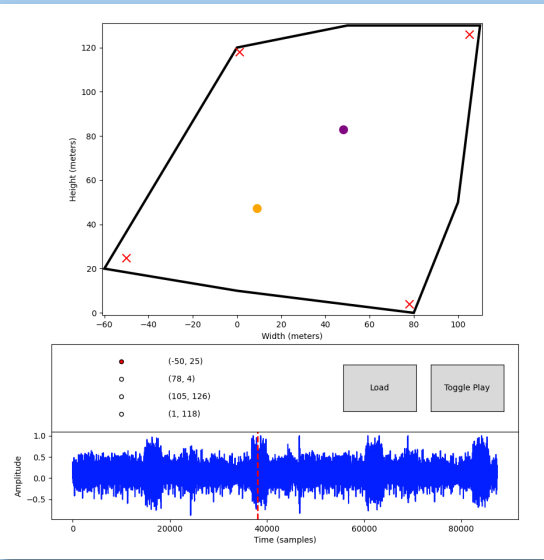
BEFORE (unprocessed sound)



AFTER (processed sound)



Localized sound

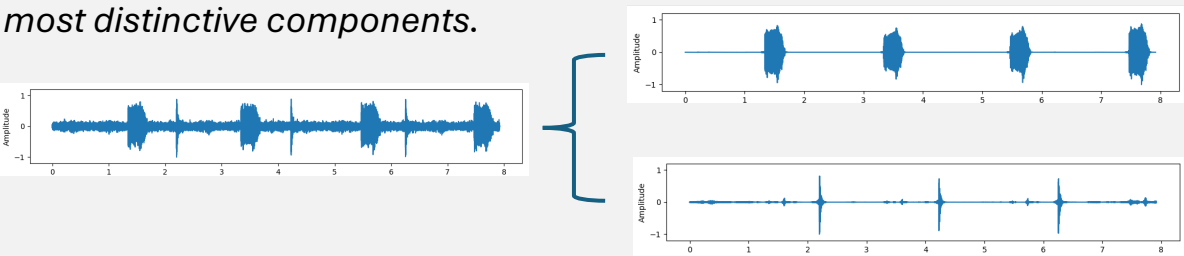


Challenges

Splitting sound signals with NMF

Explanation
The library provides an in-built method to split a single audio signal into multiple sounds by leveraging the dimensionality reduction technique of non-negative matrix factorization (NMF).

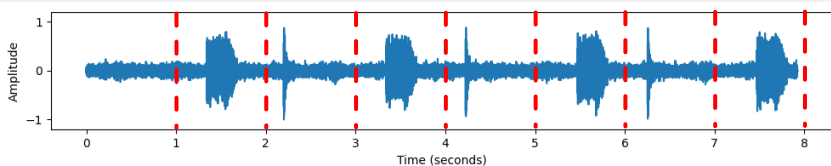
Example
A single signal is split into its two most distinctive components.



Localizing a moving sound source

Explanation
To repeatedly localize a moving sound source, the library provides a **chunking** functionality to split audio signals into smaller parts of a defined duration or sample count. All localization methods will automatically localize the sound sources of all chunks, if chunking was performed as part of the preprocessing.

Example
A single audio file chunked by duration of 1000ms (visualized).



Localization algorithms

Explanation
The library's localization functionality is based on the approach of finding the time delay of sound arriving at microphone pairs to perform multilateration by solving an optimization problem using the found time delays and known positions of the microphones.

Time delays, or time differences of arrival (TDOA), are computed using different algorithms. The library implements threshold and GCC-PHAT.
Multilateration of the sound source is a known optimization problem that takes the TDOAs between microphone pairs and the known microphone positions as inputs.

